

Python Code

```
1  """
2      This script uses the bisection method to approximate the root of a
3      scalar function.
4  """
5
6  #####
7  """
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22 """
23 #####
24
25 # import libraries
26 import numpy as np
27
28 def driver():
29
30     # use routines
31     f = lambda x: x**3 + x -4
32     a = 1
33     b = 4
34
35     # f = lambda x: np.sin(x)
36     # a = 0.1
37     # b = np.pi+0.1
38
39     tol = 1e-3
40
41     [astar,ier] = bisection(f,a,b,tol)
42     print('the approximate root is',astar)
43     print('the error message reads:',ier)
44     print('f(astar) =', f(astar))
45
46
47 # define routines
48 def bisection(f,a,b,tol):
49
50     # Inputs:
```

```
51 # f,a,b      - function and endpoints of initial interval
52 # tol - bisection stops when interval length < tol
53
54 # Returns:
55 #   astar - approximation of root
56 #   ier   - error message
57 #         - ier = 1 => Failed
58 #         - ier = 0 == success
59
60 #   first verify there is a root we can find in the interval
61
62 fa = f(a)
63 fb = f(b);
64 if (fa*fb>0):
65     ier = 1
66     astar = a
67     return [astar, ier]
68
69 # verify end points are not a root
70 if (fa == 0):
71     astar = a
72     ier = 0
73     return [astar, ier]
74
75 if (fb == 0):
76     astar = b
77     ier = 0
78     return [astar, ier]
79
80 count = 0
81 d = 0.5*(a+b)
82 while (abs(d-a)> tol):
83     fd = f(d)
84     if (fd ==0):
85         astar = d
86         ier = 0
87         return [astar, ier]
88     if (fa*fd<0):
89         b = d
90     else:
91         a = d
92         fa = fd
93     d = 0.5*(a+b)
94     count = count +1
95 #   print('abs(d-a) = ', abs(d-a))
96
97 astar = d
98 ier = 0
99 print('count = ', count)
100 return [astar, ier,]
101
102 driver()
103
104
```

105
